Goniometrix Design Roadmap

A user-driven roadmap connecting research insights to concept development and prototyping across four course phases.

Phase 1

Identify

- Problem stance
- 1:1 interviews
- User observation
- Competitive Analysis
- Opportunity Framing

Defined the need for independent, repeatable joint-angle measurement

Phase 2

Understand

- Key user groups
- Pain points + task breakdown
- Core insights
- Design requirements
- System context

Defined the need for independent, repeatable joint-angle measurement

Phase 3

Conceptualize

- 30 initial concepts
- AI-supported concept expansion
- semantic clustering
- Feasibility/novelty/ impact mapping
- Dot voting + shortlist

Defined the need for independent, repeatable joint-angle measurement

Phase 4

Realize

- System diagram + feedback loops
- Sensing → prediction → actuation
- Hardware integration
- Desktop GUI prototyping
- Final wearable prototype

Defined the need for independent, repeatable joint-angle measurement

Key Quotes

"I can't measure accurately on my own."

"I need feedback I can trust."

"I want to track progress between sessions."

Core Needs

- Independent self-measurement
- Clear, trustworthy feedback
- Repeatable tracking over time
- Low-effort, wearable setup

Design Principles

- Support independent use
- Reduce physical + cognitive load
- Make feedback actionable
- Build trust through consistency and progress tracking

Roadmap Outcome

User research and design principles guided the project from problem framing to a wearable smart goniometer with dual IMU sensing, haptic/audio feedback, and a live companion GUI.

Future direction: validate with more users, refine calibration, and expand longitudinal process tracking